

# Sepid (Sepidehsadat) Hosseini

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## Education

### Simon Fraser University

PhD in Computing Science, GPA: 4.11/4.3

- Field: 3D Computer vision and Computer Graphics.

British Columbia, Canada

May 2025

### Seoul National University

Master of Science in Electrical and Computer Engineering, GPA: 95.1/100

- Field: Intelligent Signal Processing, Computer Vision, Facial And Gesture Recognition.

Seoul, South Korea

August 2019

### Seoul National University

Bachelor of Science in Electrical and Computer Engineering, GPA: 88.6/100

- Field: Smart Road Implementation Using Arduino Car.

Seoul, South Korea

March 2017

## Work Experience

### RBC Borealis (Borealis AI)

Senior Machine Learning Researcher / Tech Lead

- Technical lead for ATOM, the Asynchronous Temporal Model, a foundation model for financial services focused on fraud detection, anomaly detection, and personalization.
- Leading research-to-production delivery of large-scale temporal modeling solutions in collaboration with research, engineering, and product teams.

Vancouver, Canada

May 2026 – Present

### RBC Borealis (Borealis AI)

Machine Learning Researcher

- Core researcher on the fraud detection team, contributing to ATOM and its adaptation to recommender systems and personalized financial applications.
- Conducted applied machine learning research leading to publications, patents, and production-oriented model development.

Vancouver, Canada

Dec 2023 – May 2026

### RBC Borealis (Borealis AI)

Machine Learning Research Intern

- Conducted research on prompt-based learning for efficient temporal domain generalization under distribution shift.

Vancouver, Canada

May 2023 – Sep 2023

### LG Electronics, CTO, Intelligent Systems Research Center

Research Intern

- Developed capsule-network-based methods for face spoofing detection and robust facial analysis.

Seoul, Korea

Jul 2017 – Sep 2017

### Samsung Techwin (Hanwha)

Software Intern

- Worked on object tracking algorithms for video analysis and surveillance-related applications.

Seoul, Korea

Jul 2015 – Aug 2015

## Selected Publications

### Improving Fairness via Noise Injection in Vision Transformers

Qiaoyue Tang, **Sepidehsadat Hosseini**, Mengyao Zhai, Thibaut Durand, Greg Mori

AFAA Workshop, ICLR 2026

### Prompting-based Efficient Temporal Domain Generalization

**Sepidehsadat Hosseini**, Mengyao Zhai, Hossein Hajimirsadegh, Frederick Tung

PUT Workshop, ICML 2025

### Radar: Fast Long-Context Decoding for Any Transformer

Yongchang Hao, Mengyao Zhai, Hossein Hajimirsadeghi, **Sepidehsadat Hosseini**, Fred Tung

ICLR 2025

### PuzzleFusion: Unleashing the Power of Diffusion Models for Spatial Puzzle Solving

**Sepidehsadat Hosseini**, Mohammad Amin Shabani, Saghar Irandoust, Yasutaka Furukawa

NeurIPS 2023 Spotlight

### Floorplan Restoration by Structure Hallucinating Transformer Cascades

**Sepidehsadat Hosseini**, Yasutaka Furukawa

BMVC 2023

## HouseDiffusion: Vector Floorplan Generation via a Diffusion Model with Discrete and Continuous Denoising

Mohammad Amin Shabani, **Sepidehsadat Hosseini**, Yasutaka Furukawa

CVPR 2023

## House-GAN++: Generative Adversarial Layout Refinement Network

Nelson Nauata, **Sepidehsadat Hosseini**, Kai-Hung Chang, Hang Chu, Chin-Yi Cheng, Yasutaka Furukawa

CVPR 2021

## GF-CapsNet: Using the Gabor Filter Responses and Capsule Networks for Age/Gender Estimation and Facial Expression Recognition

**Sepidehsadat Hosseini**, Nam Ik Cho

FG 2019

## Memberships and Volunteering Experience

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### Catch, Adapt, and Operate: Monitoring ML Models Under Drift

Rio de Janeiro, Brazil

Lead Organizer and Program Chair

ICLR 2026

- Led the organization and program committee for an ICLR workshop on monitoring, adapting, and operating machine learning models under distribution shift.

### LG Global Challenger

Seoul, Korea

Participant, Student Ambassador

2016–2017

- Selected for a fellowship program to travel across South Korea and promote Korean culture and innovation.

### Seoul National University Dormitory

Seoul, Korea

Student Advisor

2013–2014

- Supported international and domestic students in residential life and community-building activities.

### Bean Seoul

Seoul, Korea

Volunteer Instructor

2013–2017

- Volunteered in educational and recreational activities with children in orphanages.

### IUST Robotics Team

Tehran, Iran

Team Member

2009–2012

- Participated in robotics competitions as part of the university robotics research team.

## Honors and Awards

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### SFU Computing Science Graduate Fellowship

Fall 2019, Fall 2020, Fall 2021, Summer 2022, Fall 2022

### SFU Computing Science Entrance Scholarship

Fall 2019

- Award valued at 10,000 CAD.

### Seoul National University Excellent Student Global Scholarship

2017, 2018

- Award valued at \$ 8,000 per year.

### Brain Korea Scholarship, Excellent Researcher Award

2017, 2018

### LG International Student Ambassador

2016–2017

### Korean Government Scholarship

2012 - 2017

- Award valued at \$97000.